

Predicting Uncertainty Reduction in Online Alarm Flood Classification^{*}

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Abstract: Online alarm flood classification (AFC) methods support operators by assigning alarm sequences to known alarm flood classes. However, temporary class ambiguities can arise due to limited data. To address this, we extend conformal prediction-based AFC with a method to predict bifurcation points—moments when the predicted set of plausible classes becomes more specific. We use random forest regression with jackknife+ for quantile regression and introduce a delay timer to stabilize predictions. Evaluation on two datasets demonstrates that our approach has the potential to effectively forecast upcoming bifurcation points and improve prediction stability, offering decision support under uncertainty during industrial alarm floods.

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Keywords: Artificial intelligence and autonomous systems, conformal prediction, industrial alarm systems, time series classification, uncertainty quantification.

1. INTRODUCTION & RELATED WORK

Automated industrial systems depend on alarm systems to alert human operators about abnormal plant conditions requiring attention (ANSI/ISA (2016)). Alarm floods, defined by standards such as ANSI/ISA (2016) 18.2 and IEC (2022) 62682 as periods when alarm activations exceed operator-manageable limits—typically over 10 alarms within 10 minutes—pose a persistent challenge (Wang et al. (2024)). These events commonly arise due to factors including excessively configured alarms (Wang et al. (2024)) or propagating disturbances via material connections (Vogel-Heuser et al. (2015)), e.g., pipes and conveyors (Vogel-Heuser et al. (2024); Manca and Fay (2022)). Frequent occurrences of alarm floods have been observed in diverse automated industrial settings, such as continuous chemical processing, discrete manufacturing, hybrid production systems, and intralogistics facilities (Wang et al. (2024); Vogel-Heuser et al. (2015, 2024); Errington et al. (2009)). Alarm floods can severely degrade operator situational awareness (Adhitya et al. (2014)), increasing the risk of critical states being missed and potentially resulting in operational disruptions or even catastrophic accidents (EEMUA (2024)), thus underscoring the need for effective and timely operator support (Lucke et al. (2019)).

An approach to address alarm floods is the use of alarm flood classification (AFC) (Lucke et al. (2019)). AFC methods make use of historical alarm flood data, labeled through clustering methods such as those presented in Manca et al. (2022, 2021), to extract alarm patterns. These learned patterns are then matched against new, evolving alarm sequences in an online manner to assign them to

their most probable alarm flood classes (Lucke et al. (2019); Manca and Fay (2024)), thus helping operators in recognizing abnormal situations and potentially reducing their cognitive workload (Lucke et al. (2019)).

Existing AFC methods include approaches using string metrics, such as weighted dissimilarity indices (Charbonnier et al. (2015)), sequence alignment techniques, including modified Smith-Waterman (Cheng et al. (2013)) or Needleman-Wunsch algorithms (Charbonnier et al. (2016)), and feature-based techniques, such as exponentially attenuated components with nearest-neighbor classification (*EAC-1NN*) (Shang and Chen (2019)) or modified bag-of-words vectorization with logistic regression classifiers (*MBW-LR*) (Alinezhad et al. (2023)). Other AFC methods interpret alarm data as multivariate time series and use coactivation matrices classified by support vector machines (*ACM-SVM*) (Lucke et al. (2019)) or convolutional kernel-based transformations with ridge regression ensembles (*CASIM*) (Manca and Fay (2024)).

However, these AFC methods commonly struggle with temporary class ambiguities, particularly during early stages of an evolving alarm flood, where limited alarm data might hinder the differentiation between temporarily similar classes (Tamascelli et al. (2023); Manca et al. (2024)). This can result in erroneous classifications, as existing AFC methods predict only the single most probable class, despite the possibility of multiple plausible and indistinguishable classes (Manca et al. (2024)). Addressing these uncertainties has the potential to increase the trustworthiness of automation systems (Angelopoulos and Bates (2023); Thuy and Benoit (2024)).

We previously introduced a method integrating conformal prediction (CP) with early time series classification in Manca et al. (2024) to address temporary class ambiguities arising in online AFC. This method dynamically updates

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sets of plausible alarm flood classes as alarm sequences evolve, rather than providing single-point predictions. In subsequent work in Manca et al. (2025), we enhanced our method by introducing cross-CP to reduce calibration data requirements, thus improving data efficiency and applicability in data-constrained industrial environments. Despite these improvements, both methods in Manca et al. (2024) and Manca et al. (2025) remain inherently reactive, leaving operators without insights into how the model uncertainty may evolve or reduce over time, as well as the duration they should anticipate for clearer, more confident classification outcomes.

To address this, we introduce a method using random forest regression (RFR) to forecast the timing of uncertainty reduction in CP-based online AFC. Specifically, our approach estimates bifurcation points—defined as future time steps when the CP prediction sets reduce by excluding at least one previously plausible alarm flood class—and provides operators with corresponding confidence intervals. Additionally, we propose a delay timer mechanism to stabilize the CP prediction sets and reduce fluctuations. We evaluate our method using both a synthetic dataset and a dataset based on the Tennessee-Eastman process, both made publicly available, demonstrating its effectiveness across four relevant AFC methods from the literature.

2. PROPOSED METHOD

2.1 Delay Timer Mechanism

In alarm management, off-delay timers are used to prevent alarm chattering, i.e., repeated transitions between alarm states due to minor process variations or noisy measurements (EEMUA (2024)). Inspired by this practice, we introduce a delay timer to stabilize the dynamically evolving prediction sets produced by CP. Specifically, CP-based online AFC generates prediction sets Y_t^{CP} at each time step t , which can fluctuate in response to transient uncertainties, potentially undermining operator confidence in the trustworthiness of the AFC predictions.

The proposed delay timer mechanism ensures that an alarm flood class c excluded from the prediction set Y_t^{CP} at time step t is not immediately removed from the presented prediction set Y_t^{DT} . Instead, class c is only removed if it remains consistently absent from Y_t^{CP} for a predefined delay period τ_{DT} . Formally, This delay timer-stabilized prediction set is defined as follows:

$$Y_t^{\text{DT}} = \{c \mid c \in Y_t^{\text{CP}} \text{ or } \exists t' \in [t - \tau_{\text{DT}}, t) : c \in Y_{t'}^{\text{CP}}\}. \quad (1)$$

The timer duration τ_{DT} can be selected empirically or through validation to balance responsiveness and stability.

2.2 Bifurcation Point Prediction

While the delay timer stabilizes the CP prediction sets, it does not inform operators about when model uncertainty might decrease. To address this, we propose a novel approach named *Bifurcation Prediction for Alarm Flood Classification* (*BiP-AFC*). By predicting future bifurcation points, *BiP-AFC* assists operators in anticipating

when less ambiguous classifications will likely become available. Consequently, they may decide to postpone or initiate corrective actions at this time.

Given the CP-based prediction set $Y_t^{\text{CP}} \subseteq \mathcal{C}$ at the time step t , where $\mathcal{C}$ is the set of all possible alarm flood classes, we define the next bifurcation point t_{BP} as the earliest future time at which the prediction set size decreases:

$$t_{\text{BP}} = \min \{t' > t : |Y_{t'}^{\text{CP}}| < |Y_t^{\text{CP}}|\}. \quad (2)$$

To estimate t_{BP} , *BiP-AFC* employs an RFR model, which was specifically selected due to its robustness against noisy data (Sage (2018)), capability to handle nonlinear relationships (Sage (2018)), and suitability for quantile regression (Foygel Barber et al. (2021)). First, an AFC model, which outputs either posterior class probabilities or class-specific distance measures, is trained on historical alarm floods. Subsequently, CP is calibrated on a dedicated calibration dataset (Manca et al. (2024)) or using cross-CP (Manca et al. (2025)). To prevent overfitting, *BiP-AFC*'s RFR model is trained separately on data not utilized in the AFC model training or CP calibration. Historical bifurcation times are calculated from this independent data.

The RFR model is trained using an input feature vector $\mathbf{x}_t$, selected through preliminary testing to encode information regarding temporal dynamics, recent bifurcation history, model uncertainty, and current class ambiguity. That is, $\mathbf{x}_t$ at time t comprises the following elements:

- *Relative current time*: Captures the progression of the evolving alarm flood, normalized against the maximum alarm flood duration in the training data.
- *Relative time since last bifurcation*: Represents the elapsed normalized duration since the most recent bifurcation point, indicating how long the prediction set size has remained unchanged.
- *Binary one-hot vector of classes in the CP prediction set*: Encodes the currently plausible alarm flood classes given by CP, incorporating class ambiguity.
- *Current class probabilities or distances from the AFC model*: Offers a numeric representation of the relative confidence or proximity of each alarm flood class at the current time, reflecting evolving model certainty.

Given this feature vector $\mathbf{x}_t$, the RFR model outputs an estimate of the next bifurcation point $\hat{t}_{\text{BP}}$:

$$\hat{t}_{\text{BP}} = f_{\text{RFR}}(\mathbf{x}_t). \quad (3)$$

Furthermore, to provide operators with reliable uncertainty quantification, we apply the Jackknife+ CP method (Foygel Barber et al. (2021)) to obtain statistically rigorous confidence intervals for the predicted bifurcation points. Jackknife+ generates valid confidence intervals by systematically excluding subsets of the training data to quantify prediction uncertainty (Foygel Barber et al. (2021)). Formally, given a significance level $\alpha \in (0, 1)$, we derive a confidence interval:

$$\mathcal{I}_t^{(1-\alpha)} = [\hat{t}_{\text{BP}}^{\text{lower}}, \hat{t}_{\text{BP}}^{\text{upper}}], \quad (4)$$

such that the true bifurcation point t_{BP} lies within this interval with probability $1 - \alpha$.

Finally, as a practical post-processing step, we ensure that the computed confidence intervals are consistent with the online classification scenario. That is, we adjust any confidence interval whose lower bound is less than the next time step, since predicting bifurcation points earlier than the next observation or in the past is impossible. Thus, the adjusted confidence interval is computed as follows:

$$\mathcal{I}_{t,\text{adj}}^{(1-\alpha)} = [\max(\hat{t}_{\text{BP}}^{\text{lower}}, t + 1), \hat{t}_{\text{BP}}^{\text{upper}}]. \quad (5)$$

In practice, *BiP-AFC* can be integrated into an operator-friendly interface, displaying online predictions of AFC bifurcation points along with confidence intervals. Operators can adjust the significance level α according to their preferences, or default intervals can be shown.

3. EVALUATION & DISCUSSION

3.1 Synthetic Evaluation Dataset

To evaluate our method, we use a synthetic dataset designed to simulate alarm floods in industrial contexts. The dataset comprises labeled alarm flood sequences reflecting various degrees of temporary class ambiguities, generated according to procedures introduced in prior works (Manca et al. (2024, 2025)). We chose synthetic data because publicly available alarm datasets with adequate size and clearly defined ground truth are scarce (Chioua et al. (2019)), and synthetic data enables a controlled evaluation of how AFC models handle evolving uncertainty over time.

The dataset includes 10 alarm variables and five different alarm flood classes. Differences between classes arise from varying alarm activation probability patterns assigned to these alarm variables. Temporary class ambiguities are embedded into the alarm data generation process by configuring overlapping activation patterns among multiple classes. As a result, several classes share common initial alarm activations, making them temporarily indistinguishable to AFC models. Over time, diverging patterns emerge, i.e., certain alarms activate only in specific classes, allowing models to eventually resolve ambiguities. These bifurcation points differ among samples due to probability-driven alarm activations (Manca et al. (2024)).

For our experiments, we generated 1875 synthetic alarm flood samples, each with a duration of 60 minutes at a sampling interval of 1/min. More detailed information regarding the dataset generation and characteristics can be found in Manca et al. (2024).

3.2 Tennessee-Eastman Process Evaluation Dataset

We further evaluate our method using a dataset generated from the well-established Tennessee-Eastman Process (TEP) simulation model (Downs and Vogel (1993); Bathelt et al. (2015)), an open benchmark widely used in process monitoring and fault diagnosis research (Melo et al. (2022)). The TEP includes six interconnected process modules such as a two-phase chemical reactor and a vapor-liquid separator. A detailed description of the plant, control system, and process variables is provided in Downs and Vogel (1993), Ricker (1996), and Bathelt et al. (2015).

For our AFC task, we focus on the 22 process variables listed in Table 4 of Downs and Vogel (1993) and variable 23 from Table 5 of Downs and Vogel (1993), as well as variables 48 and 49 from Table 1 of Bathelt et al. (2015), which capture relevant process state information. To generate alarms from the continuous process data, we apply high and low alarm thresholds at ± 2 standard deviations from the normal steady-state mean, and implement a deadband of ± 1.5 standard deviations for alarm deactivation. This hysteresis reduces chattering and more accurately simulates real-world alarm systems (EEMUA (2024)).

We simulate five fault scenarios: abrupt composition changes in feed stream 4 (faults 1 and 2), abrupt feed loss in stream 1 (fault 6), an oscillating sticking valve in the reactor cooling system (fault 14), and a combined scenario involving fault 1 and a step change in condenser cooling water temperature (fault 5). Each scenario is simulated 200 times, resulting in 1,000 samples. Simulations are run for one hour at a sampling rate of 1/min. Fault severity is randomized between 60% and 100% to capture variations in process dynamics (Bathelt et al. (2015)).

3.3 Experimental Setup

We evaluate our method using four relevant AFC methods from the literature: *ACM-SVM* (Lucke et al. (2019)), *EAC-1NN* (Shang and Chen (2019)), *MBW-LR* (Alinezhad et al. (2023)), and *CASIM* (Manca and Fay (2024)). The hyperparameters for these classifiers follow the configurations reported in prior work (Manca et al. (2024)). For uncertainty quantification, we adopt the CP method introduced in Manca et al. (2024), using a significance level of $\alpha = 0.05$, as this setting demonstrated favorable performance in earlier evaluations.

To analyze the effect of the proposed delay timer, we evaluate delay durations ranging from 0 to 15 minutes. For our proposed *BiP-AFC* method, we use the implementation and default parameters of the RFR from the `scikit-learn` library (Pedregosa et al. (2011)) and the Jackknife+ CP from the `MAPIE` library (Cordier et al. (2023)). We evaluate prediction interval performance across significance levels $\alpha \in [0.01, 0.30]$ in steps of 0.01, as lower confidence levels are impractical for industrial use and the chosen resolution allows for fine-grained analysis.

The evaluation employs 5-fold stratified cross-validation. For the synthetic dataset, in each fold, 100 samples per class are allocated to AFC model training, another 100 samples per class for calibrating our CP method from (Manca et al. (2024)), an additional 100 samples per class for training the novel *BiP-AFC* method, and finally 75 samples per class for testing. For the TEP dataset, the split is 60 samples per class for training, 50 for calibration, 50 for *BiP-AFC* training, and 40 for testing.

3.4 Evaluation Results for the Synthetic Dataset

Fig. 1 illustrates the impact of varying delay timer durations on the performance of the four AFC methods. *ACM-SVM*, *MBW-LR*, and *CASIM* initially exhibit relatively high average numbers of prediction set changes (Fig. 1(a)). Increasing the delay timer notably reduces these fluctuations, especially for *CASIM*, which shows a

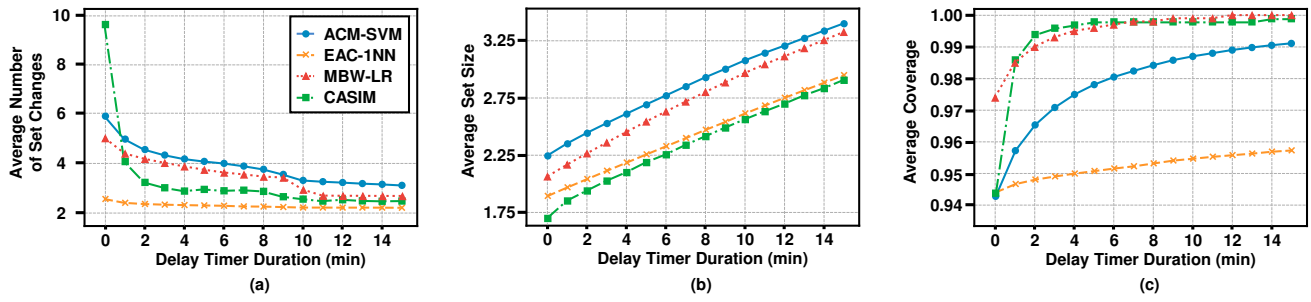


Fig. 1. Impact of varying delay timer durations on the performance of the examined alarm flood classification methods for the synthetic dataset. (a) Average number of prediction set changes, (b) Average prediction set size, and (c) Average prediction set coverage, evaluated over five folds per setting.

significant drop already at a delay duration of 1 minute, suggesting frequent class additions and removals within single time steps. Concurrently, the average prediction set size moderately increases with longer delay durations (Fig. 1(b)). Additionally, prediction coverage, i.e., the percentage of times the true class is included in the predicted set, increases for all methods with longer delay durations, particularly for *MBW-LR* and *CASIM* (Fig. 1(c)). Based on these observations, a delay timer duration of three minutes is selected for subsequent evaluations of *BiP-AFC*.

Table 1 summarizes the mean number of bifurcation points in the training and test data across the four AFC methods. *ACM-SVM* has the highest number of bifurcation points, followed by *MBW-LR*, *CASIM*, and *EAC-1NN*, reflecting differences in their handling of evolving class ambiguities.

Table 2 presents the mean absolute error (MAE) between the observed coverage, i.e., the percentage of times the true bifurcation point lies within the predicted confidence interval, and the expected confidence level ($1 - \alpha$). The low MAE values for all AFC methods indicate that *BiP-AFC* reliably quantifies prediction uncertainty. Additionally, Table 2 shows the MAE between the true bifurcation points and the RFR's point predictions (without quantile regression). All four AFC methods exhibit relatively low errors, with *EAC-1NN*, *MBW-LR*, and *CASIM* showing average MAEs below one minute.

Fig. 2 shows the average widths of prediction intervals produced by *BiP-AFC* across varying α values. For all AFC methods, the width of prediction intervals decreases consistently as the significance level α increases, demonstrating expected behavior—narrower intervals correspond to lower confidence. *CASIM*, *MBW-LR*, and *EAC-1NN* produce considerably narrower prediction intervals than *ACM-SVM*, maintaining interval widths under four minutes for the majority of α values, indicating a reasonable balance between coverage and interval width.

3.5 Evaluation Results for the Tennessee-Eastman Process Dataset

Fig. 3 illustrates the impact of varying delay timer durations on AFC prediction stability for the TEP dataset. Similar to the synthetic data, the delay timer effectively reduces the number of prediction set changes across all methods, particularly for *CASIM* and *MBW-LR* (Fig. 3(a)). Increasing the delay also leads to moderately larger prediction sets (Fig. 3(b)) and higher coverage values (Fig. 3(c)).

Table 1. Mean and standard deviation of the number of bifurcation points for the synthetic dataset

Classifier	Training	Test
ACM-SVM	1853.40 $\pm$ 11.88	1395.60 $\pm$ 12.19
EAC-1NN	1094.20 $\pm$ 31.03	814.00 $\pm$ 23.81
MBW-LR	1747.60 $\pm$ 118.04	1311.20 $\pm$ 87.15
CASIM	1425.00 $\pm$ 82.88	1065.60 $\pm$ 48.42

Table 2. Mean Absolute Errors (MAE) between observed coverage and expected confidence levels ($1 - \alpha$), and between predicted and true bifurcation points for the synthetic dataset

Classifier	MAE Coverage	MAE Points
ACM-SVM	0.0145 $\pm$ 0.0079	1.2930 $\pm$ 2.3687
EAC-1NN	0.0156 $\pm$ 0.0066	0.5240 $\pm$ 1.2717
MBW-LR	0.0126 $\pm$ 0.0026	0.6475 $\pm$ 0.8219
CASIM	0.0111 $\pm$ 0.0067	0.7127 $\pm$ 1.5700

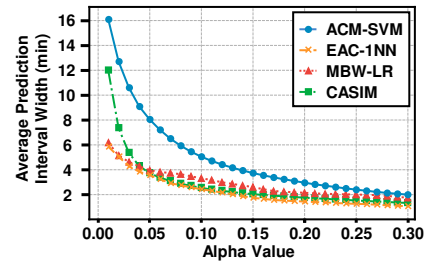


Fig. 2. Comparison of average prediction interval widths across different alpha values for the examined methods, calculated over five folds per setting for the synthetic dataset.

Based on these trends, we again use a delay duration of three minutes for subsequent *BiP-AFC* evaluations.

Table 3 shows that *ACM-SVM* exhibits the highest number of bifurcation points in both training and test data, followed by *MBW-LR*, *CASIM*, and *EAC-1NN*. This ranking is consistent with the one observed in the synthetic data.

Table 4 presents the MAE between the observed coverage and the target confidence level $1 - \alpha$. While all methods maintain relatively low errors, *CASIM* and *MBW-LR* exhibit the most accurate uncertainty quantification. Regarding the accuracy of point predictions, *CASIM* and *MBW-LR* again achieve lower MAE values, indicating more precise estimation of upcoming bifurcation points.

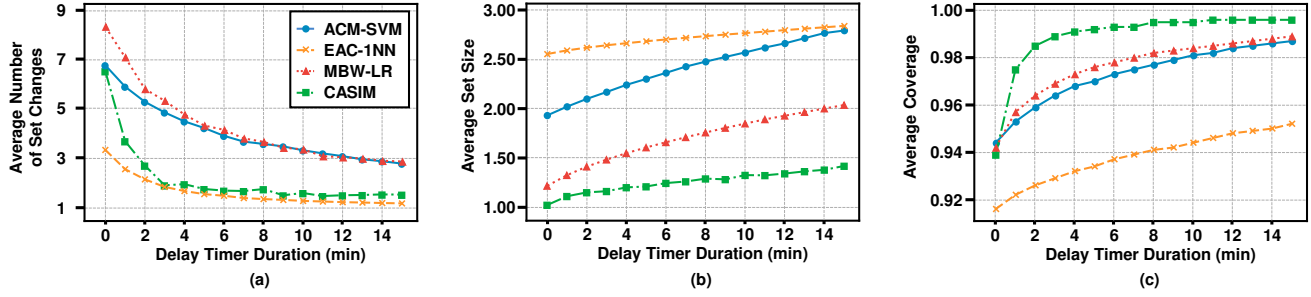


Fig. 3. Impact of varying delay timer durations on the performance of the examined alarm flood classification methods for the Tennessee-Eastman process dataset. (a) Average number of prediction set changes, (b) Average prediction set size, and (c) Average prediction set coverage, evaluated over five folds per setting.

These results demonstrate that *BiP-AFC* is capable of predicting bifurcation points with only a few minutes deviation, even in complex alarm flood scenarios involving varying disturbance severities and propagation dynamics.

Fig. 4 displays the average widths of the predicted confidence intervals across α values. All methods show consistent behavior, with interval widths narrowing as α increases. Compared to the synthetic dataset, the intervals are wider, which may be attributed to the increased complexity of the alarm flood scenarios in the TEP. In particular, varying disturbance severities and propagation dynamics introduce greater uncertainty, requiring broader prediction intervals to maintain valid coverage.

On a computer with an Intel(R) Core(TM) Ultra 9 185H processor (2.3 GHz), we measured an average per-step inference latency using *BiP-AFC* of 4.2 ± 0.9 ms.

3.6 Discussion

The proposed *BiP-AFC* method improves CP-based AFC by introducing the capability to predict when uncertainty is likely to reduce. This allows human operators, for the first time, not only to recognize the presence of AFC ambiguity but also to anticipate when more definitive class assignments can be expected. Such predictive insights have the potential to support more informed and timely operator decisions during evolving alarm floods.

Additionally, the introduced delay timer mechanism effectively stabilizes fluctuating prediction sets. Our evaluation further demonstrates that *BiP-AFC* can yield relatively narrow prediction intervals for upcoming bifurcation points. However, the precision of these intervals strongly depends on the chosen AFC method and the ambiguity of the underlying alarm patterns.

4. CONCLUSIONS & FUTURE WORK

We introduced *BiP-AFC*, a novel method for predicting bifurcation points to manage evolving uncertainty in CP-based AFC. Our approach enables operators to anticipate when classification ambiguity is likely to reduce, thus supporting more informed decision-making under uncertainty. Additionally, we proposed a delay timer mechanism that improves prediction stability by reducing fluctuations in the predicted class sets.

Evaluation on both a synthetic dataset and a dataset derived from the TEP demonstrated the effectiveness of

Table 3. Mean and standard deviation of the number of bifurcation points for the Tennessee-Eastman process dataset

Classifier	Training	Test
ACM-SVM	603.60 $\pm$ 58.02	471.80 $\pm$ 51.92
EAC-1NN	227.00 $\pm$ 30.46	178.00 $\pm$ 17.24
MBW-LR	460.60 $\pm$ 122.01	367.40 $\pm$ 78.66
CASIM	299.80 $\pm$ 32.49	233.40 $\pm$ 27.60

Table 4. Mean Absolute Errors (MAE) between observed coverage and expected confidence levels ($1 - \alpha$), and between predicted and true bifurcation points for the Tennessee-Eastman process dataset

Classifier	MAE Coverage	MAE Points
ACM-SVM	0.1004 $\pm$ 0.0069	3.5827 $\pm$ 4.7332
EAC-1NN	0.2142 $\pm$ 0.0284	4.0613 $\pm$ 4.8142
MBW-LR	0.0228 $\pm$ 0.0095	2.0768 $\pm$ 3.0343
CASIM	0.0104 $\pm$ 0.0056	2.3272 $\pm$ 2.2046

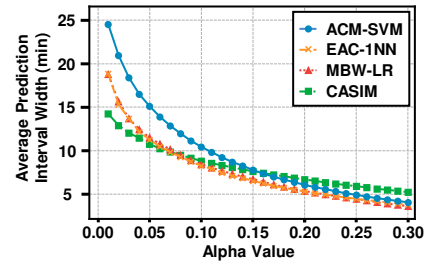


Fig. 4. Comparison of average prediction interval widths across different alpha values for the examined methods, calculated over five folds per setting for the Tennessee-Eastman process dataset.

our approach across four AFC methods. *BiP-AFC* reliably predicted upcoming bifurcation points and provided well-calibrated confidence intervals, even in complex alarm flood scenarios with varying disturbance severities and propagation dynamics.

One advantage of the method lies in its model-agnostic design: it can be applied to any AFC method that integrates CP and outputs class-specific probabilities or distance measures. However, the method does not currently provide insight into the qualitative nature of the predicted bifurcation points. Specifically, it does not indicate which alarm flood classes are most likely to be excluded or retained, nor does it relate upcoming bifurcations to expected alarm

patterns or required input conditions. Future work should address these aspects to further improve operator decision support and interpretability. Moreover, future work could explore novel deep learning-based AFC solutions that leverage regularization and calibration techniques to directly estimate and improve uncertainty quantification.

DATA & CODE AVAILABILITY

All evaluation datasets, code for *BiP-AFC* and the delay timer, reproducible 5-fold experiments, and configurations for all AFC and CP methods are publicly available at DOI: 10.24433/CO.3008979.v3.

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